

ALPENA COUNTY REGIONAL, MI, USA

WMO: 726390

Lat: 45.072N

Lon: 83.564W

Elev: 684

StdP: 14.34

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94849

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.7	-0.1	-11.5	3.0	-4.2	-6.5	4.0	1.1	21.6	20.7	20.0	21.5	3.9	270	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	88.0	71.1	84.6	69.5	81.5	67.9	74.1	83.6	72.1	80.8	70.2	78.0	10.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.9	116.8	78.9	69.1	109.7	76.8	67.4	103.4	74.6	37.9	84.0	36.2	80.9	34.6	78.0	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	20.0	18.0	16.3	DB	-14.3	93.7	7.2	3.2	-19.5	96.0	-23.7	97.8	-27.8	99.6	-33.0	101.9
(5)				WB	-14.7	77.8	7.1	2.4	-19.8	79.5	-23.9	80.9	-27.9	82.2	-33.0	83.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.3	19.5	20.4	29.1	41.0	53.1	63.1	68.0	66.4	59.4	47.7	36.4	26.4	
	DBStd	19.21	10.85	10.81	10.85	8.62	8.46	7.26	5.90	5.84	7.63	8.38	8.58	9.40	
	HDD50	4085	945	830	654	292	59	1	0	0	9	149	415	731	
	HDD65	7883	1410	1250	1114	719	383	122	36	53	201	541	858	1196	
	CDD50	2018	0	0	6	23	155	393	558	507	293	77	6	0	
	CDD65	343	0	0	0	1	15	64	129	96	34	4	0	0	
	CDH74	3408	0	0	6	26	222	733	1240	817	323	41	0	0	
	CDH80	984	0	0	1	5	62	247	374	210	78	7	0	0	
Wind	WSAvg	7.3	8.0	8.1	8.1	8.6	7.7	6.4	6.1	6.0	6.3	7.2	7.7	7.7	
Precipitation	PrecAvg	29.6	1.9	1.5	1.7	2.8	2.9	2.8	3.3	3.1	2.9	2.8	2.1	1.9	
	PrecMax	36.0	4.4	3.5	6.2	6.4	6.6	5.5	5.8	6.7	6.5	6.4	4.4	4.5	
	PrecMin	23.3	0.5	0.5	0.3	1.2	1.1	0.2	1.1	1.3	0.7	0.7	0.8	0.3	
	PrecStd	3.4	0.9	0.7	1.3	1.2	1.3	1.3	1.2	1.5	1.4	1.3	1.0	1.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.9	52.1	69.2	77.7	86.4	91.4	91.7	90.4	87.2	79.6	65.1	54.3	
		MCWB	43.2	46.1	56.1	59.7	66.6	73.6	73.6	72.8	71.1	65.2	55.1	49.7	
	2%	DB	41.5	43.2	56.6	68.3	80.6	86.6	87.8	85.2	81.8	72.1	58.5	46.8	
		MCWB	38.4	38.3	48.9	54.2	64.2	69.8	71.7	70.6	68.3	62.0	51.7	43.0	
	5%	DB	37.1	38.1	49.7	61.6	74.5	82.2	84.1	81.8	77.5	66.7	53.5	41.9	
		MCWB	35.1	34.6	42.3	50.6	60.6	67.9	69.9	68.9	66.1	59.4	48.3	39.0	
	10%	DB	34.4	35.0	43.7	56.0	69.3	77.6	80.9	78.6	73.5	61.9	49.4	38.5	
		MCWB	32.9	32.6	37.9	46.2	58.0	64.9	67.9	67.0	64.2	55.6	45.1	36.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.4	46.5	58.6	62.0	70.3	75.5	77.4	75.6	73.4	68.7	57.8	51.2
			MCDB	46.0	50.4	66.9	72.5	82.3	88.4	87.9	84.2	83.2	74.9	62.7	53.4
2%		WB	38.5	38.9	50.2	56.5	66.5	72.4	74.5	73.0	70.4	63.6	53.1	43.7	
		MCDB	40.6	42.2	56.5	66.6	76.5	83.1	83.5	81.6	77.8	69.9	57.5	46.4	
5%		WB	35.3	35.1	43.2	51.4	63.0	69.7	72.2	71.1	68.3	59.9	49.0	39.4	
		MCDB	36.9	37.5	48.7	59.6	72.4	78.5	80.5	78.5	74.9	66.1	53.3	41.6	
10%		WB	32.9	32.8	38.6	46.9	59.2	67.2	70.2	69.2	65.8	56.5	45.6	36.2	
		MCDB	34.5	35.0	43.0	55.0	67.6	74.9	77.7	75.4	72.1	61.2	49.1	38.4	
Mean Daily Temperature Range	5% DB	MDBR	14.4	17.1	18.3	19.9	22.3	22.6	22.1	21.5	21.4	18.1	14.1	12.4	
		MCDBR	14.4	18.4	25.1	30.6	30.2	28.3	26.3	24.9	26.7	26.3	21.1	16.3	
		MCWBR	12.6	14.6	17.0	18.9	16.2	14.4	12.8	12.5	14.6	16.9	15.9	13.6	
	5% WB	MCDBR	13.7	17.5	23.7	27.9	27.2	24.8	22.3	21.8	22.2	23.9	19.8	15.7	
		MCWBR	12.2	14.5	17.3	18.8	16.0	14.1	12.3	12.2	13.5	16.6	16.0	13.7	
Clear-Sky Solar Irradiance	taub	0.285	0.297	0.308	0.350	0.371	0.403	0.402	0.390	0.368	0.338	0.307	0.280		
	taud	2.415	2.355	2.444	2.371	2.386	2.318	2.341	2.370	2.427	2.492	2.524	2.480		
	Ebn at Noon	265	282	295	289	285	275	273	273	270	264	256	256		
	Edh at Noon	23	29	31	36	37	40	38	36	31	26	21	19		
All-Sky Solar Radiation	RadAvg	398	689	1168	1472	1750	1912	1932	1640	1281	766	424	292		
	RadStd	36	76	114	133	159	139	74	77	117	82	58	30		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page